A margin in Weil's criterion

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Abstract

Let $K(r) = \sum_\rho r^{\rho-1/2}/(\rho(1-\rho))$, summed over the nontrivial zeros of ζ, and $G_N = (\sqrt{mn}\,K(m/n))_{m,n\le N}$, whose entries can be computed from the primes. The Riemann hypothesis holds exactly when every G_N is positive semidefinite, a discrete analogue of a theorem of Matsumoto and Suzuki, and G_N is then the Gram matrix of $w_n = (n^\rho/|\rho|)_\rho$: $x^T G_N x = \sum_\rho |P(\rho)|^2/|\rho|^2$ for the Dirichlet polynomial $P(s) = \sum_n x_n n^s$. We prove that, under the hypothesis, $x^T G_N x \ge 0.0577\, x_N^2$ for every $N \ge 2$ and every real x: this sum over the zeros is at least 0.0577 times the square of the top coefficient of P, uniformly in its length. Since 0.0577 exceeds $G(1,1) = 2 + \gamma - \log 4\pi = 0.0462$, every Gram–Schmidt pivot of $w_1, w_2, \dots$ is at least the first, and $\pm w_1$ are the only shortest nonzero vectors of every lattice $\mathbb{Z}w_1 + \cdots + \mathbb{Z}w_N$. Stated for G_N, each of these properties is equivalent to the Riemann hypothesis. The proof picks a band-limited test function that makes the prime side of the explicit formula exactly Landau's quadratic form, then bounds that form by Schur's test and the archimedean term by an interval certificate. It is formalised in Lean 4 with Mathlib, on Lean's standard axioms alone. Numerically, the pivots exceed the first because zeros are scarce at low height: for a constant density of zeros every pivot equals the first, while the same density with no zeros below height 2π makes the second pivot 1.96 times the first. The pivot property fails for every curve over a finite field, where the Riemann hypothesis is a theorem, and both properties fail for a positive proportion of real Dirichlet characters.

1 Introduction

Let ρ run over the nontrivial zeros of the Riemann zeta function, counted with multiplicity. Weil's positivity criterion [1] says that the Riemann hypothesis (RH) holds if and only if a quadratic functional, built from the

primes and the archimedean place through the explicit formula, is nonnegative on a suitable space of test functions. This paper restricts it to one orbit. For $r > 0$ let

$$K(r) = \sum_\rho \frac{r^{\rho-1/2}}{\rho(1-\rho)}, \qquad G(m,n) = \sqrt{mn}\, K(m/n), \tag{1}$$

and let $G_N = (G(m,n))_{m,n\le N}$ and $Q_N(x) = \sum_{m,n\le N} x_m x_n G(m,n)$ for real x. The series converges absolutely, $K(1/r) = K(r)$, and

$$d_1 := K(1) = \sum_\rho \frac{1}{\rho(1-\rho)} = 2 + \gamma - \log 4\pi = 0.04619\ldots, \tag{2}$$

γ being Euler's constant. Through the explicit formula K is a function of Chebyshev's ψ (Proposition 2.1), so G_N can be computed from the primes without the zeros. Under RH, with $\rho = \frac12 + i\gamma_\rho$, $G(m,n) = \sum_\rho m^\rho \overline{n^\rho}/|\rho|^2$: G_N is the Gram matrix of the vectors $w_n = (n^\rho/|\rho|)_\rho$ in ℓ^2 of the zeros, and $Q_N(x) = \sum_\rho |\sum_{n\le N} x_n n^\rho|^2/|\rho|^2$. Conversely, RH holds if and only if every Q_N is nonnegative at the integer vectors (Proposition 2.2). That positivity layer is known in substance: K is the function H_1 of Matsumoto and Suzuki, who prove that RH holds exactly when $H_1(e^t) - H_1(1)$ is a screw function on $\mathbb{R}$ [2, Theorem 1.3], in Kreĭn and Langer's sense [3]: a continuous g with $g(-t) = \overline{g(t)}$ whose kernel $g(t-u) - g(t) - g(-u) + g(0)$ is nonnegative definite. The 2×2 minors of the G_N are nonnegative exactly when $|K(r)| \le d_1$ at the rationals, which under RH is the triangle inequality, as Nicolas notes [4]; the lower bound $K(r) \ge -d_1$ alone is already equivalent to RH (Proposition 2.2).

We ask for more than positivity: a margin. Let δ_n be the n-th unit vector, and let d_k be the k-th Gram–Schmidt pivot of the orbit, the infimum of $Q_k(\delta_k - \sum_{n<k} c_n\delta_n)$ over real c: under RH, the squared distance of w_k from the span of $w_1,\ldots,w_{k-1}$. The main result bounds $Q_N(x) = \sum_\rho |P(\rho)|^2/|\rho|^2$, for the Dirichlet polynomial $P(s) = \sum_{n\le N} x_n n^s$, below by a constant times the square of its top coefficient, uniformly in N.

Theorem 1.1. *Assume RH. For every $N \ge 2$ and every real x,*

$$Q_N(x) \ge c_0\, x_N^2, \qquad c_0 = \frac{999}{1000}\cdot\frac{2}{45}\cdot\frac{13}{10} = 0.05772.$$

Hence $d_k \ge c_0 > 1.249\, d_1$ for every $k \ge 2$; every integer vector with a nonzero coordinate at some $n \ge 2$ has $Q_N(a) \ge c_0$; and $\pm\delta_1$ are the only nonzero integer vectors with $Q_N(a) \le d_1$.

As $c_0 > d_1$, the margin clears the first pivot and the first vector. Consider

(A) $d_k \geq d_1$ for every $k \geq 1$ (the pivot form);

(B) $Q_N(a) \geq d_1$ for every N and every nonzero $a \in \mathbb{Z}^N$ (the shortest-vector form): under RH, w_1 is a shortest nonzero vector of the lattice spanned by $w_1, \ldots, w_N$.

Corollary 1.2. *The following are equivalent:* (i) *the Riemann hypothesis;* (ii) (A)*;* (iii) (B)*.*

The implications from (A) and from (B) to RH are formal: each makes every Q_N nonnegative at the integer vectors, which is RH by Proposition 2.2. The converse is Theorem 1.1. The threshold d_1 is a choice: given Theorem 1.1, $d_k \geq \varepsilon$ for every k is equivalent to RH for each $\varepsilon \in (0, d_1]$, and $\varepsilon = d_1$ is the value at which the statement becomes a lattice minimum.

The true values are larger. Numerically $d_2/d_1 = 1.7184\ldots$, certified in ball arithmetic, and the second minimum of the lattice, the least $Q_N(a)/d_1$ over integer $a \neq 0, \pm\delta_1$, is $1.9387\ldots$ for every $2 \leq N \leq 10^4$, at $a = \delta_1 + \delta_2$ (Section 3.1). The pivots grow, numerically, to $d_{500} = 74.9\, d_1$, and they are at most $\log(k/2\pi) + 1 - \gamma + o(1)$ (Section 3). So Theorem 1.1 is far from sharp beyond the first few k; its point is that it is uniform.

In the flat model, where the zeros are replaced by a constant density, normalised so that $K(1) = d_1$, the Gram matrix is $d_1 \min(m, n)$ and every pivot equals d_1 (Proposition 2.5). Numerically, the pivots exceed d_1 because zeros are scarce at low height, where the weight $1/(\frac{1}{4} + t^2)$ of the sum $K(1)$ is concentrated. With a constant density that vanishes below height 2π, $d_k/d_1 = 1.96$, 2.78 and 12.9 at $k = 2$, 3 and 30. With the zeros' own density, $(1/2\pi)\log(t/2\pi)$ above height 2π, they are 2.00, 3.00 and 24.8, and 2.00, 2.99 and 25.4 if that density starts only at the first zero, at height 14.13. A density with no gap that is merely small near height 0, $(1/2\pi)\log(1 + t/2\pi)$, gives 1.94, 2.73 and 10.7. But $\max(1, \log(t/2\pi))/2\pi$, which grows like the zeros' density and is constant below height $2\pi e$, gives only 1.04, 1.07 and 1.97, as 93% of its weight lies below height 2π (`scripts/ball_orbit.py --densities`). So at $k = 2$, where the margin is decided, what counts is how little density lies below height about 2π; the growth of the density matters only for larger k. The zeros themselves give 1.72, 2.37 and 14.6: their discreteness pulls the pivots down from the density's values, and Theorem 1.1 says that under RH it never pulls them down to d_1.

As far as our literature search reaches, no source bounds $\sum_\rho |P(\rho)|^2/|\rho|^2$ below by the top coefficient of P uniformly in its length, states (A) or (B),

or states RH through a lattice minimum; Section 7 places the paper in the literature. We claim Theorem 1.1 and the corollary it gives; the positivity layer, the kernel and the constant d_1 are not ours.

The analogues

The same construction can be made for a curve over a finite field, where RH is a theorem, and for a Dirichlet L-function, where GRH is expected (Section 6). For a curve the zeros repeat vertically, so their density is constant, and the Frobenius forces an integral relation among the orbit vectors, with the coefficients of its minimal polynomial: a pivot vanishes, and (A) fails for every curve. The relation is the zero vector of the lattice the orbit spans, so it does not touch the lattice form (B′) of Definition 2.4; but (B′) fails for 43 of the 148 curves we sampled, at $N = 2$, and the others pass on the indices we tested. On the orbit of the Frobenius graphs, whose integer span is the lattice $\mathbb{Z}[z]/(m)$ for the minimal polynomial m of Frobenius, (B′) holds, by integrality, whenever m is irreducible. For real characters the two forms coincide at $k = 2$, where a positive proportion of them fail; for complex characters they part, through $\operatorname{Im} L'/L(1,\chi)$. For ζ no integral structure is known, and (B) rests on the analytic margin alone. So, as far as these tests reach, the margin is a property of ζ, of $d_1 = 0.046$ and of the scarcity of its zeros at low height, none below 14.13, not of the construction; and Corollary 1.2 is a form of RH, not a route to it.

The proof and its formalisation

The proof of Theorem 1.1 (Section 4) applies Weil's explicit formula to $h(t) = |D(t)|^2\varphi(t/N)/N^2$, with $D(t) = \sum_{n\le N} x_n\sqrt{n}\,n^{-it}$ and a fixed test function φ whose Fourier transform is supported in $[-\frac{4}{5},\frac{4}{5}]$. Distinct frequencies $\log(jn)$ and $\log j'$ are at least $4/(5N)$ apart, so the prime side is exactly Landau's quadratic form $2\sum_n \Lambda(n)n^{-1/2}\sum_{jn\le N} c_j c_{jn}$, $c_n = x_n\sqrt{n}$: the off-diagonal term that Landau's formula [13], in Fujii's sharpening under RH [15] and Gonek's uniform version [14], gives the mean value of $|D|^2$ over the zeros. A minorant of the archimedean term, band-limited to $[-1,1]$, keeps only the diagonal, and an interval certificate checks it against the digamma function. The last step is arithmetic: under the diagonal $(\log N + \frac{17}{20})\sum_n c_n^2$, Landau's form leaves $\frac{13}{10}c_N^2$ to spare at the top vertex N of the divisibility graph, by Schur's test with an explicit bound for Mertens' sums (Proposition 4.8). The whole proof is formalised in Lean 4 [48] with Mathlib [49], in the repository of the paper [50], and depends only on the standard axioms

(Section 5). Appendix A names the Lean theorem behind each numbered result. Remarks, conjectures and numerical statements are not Lean theorems.

Organisation

Section 2 sets up the kernel, the orbit and the forms. Section 3 derives Corollary 1.2 and its lattice and window forms from Theorem 1.1 and bounds the pivots above, and Section 4 proves Theorem 1.1. Section 5 describes the formalisation and Section 6 the analogues. Section 7 discusses the literature, and Section 8 the smallest eigenvalue of G_N and some questions.

2 The kernel, the orbit and the forms

Proposition 2.1. *The series* (1) *converges absolutely for every $r > 0$, $K(1/r) = K(r)$, $K(1) = 2 + \gamma - \log 4\pi$, and for $x \geq 1$, with $u = \log x$,*

$$K(x) = \cosh(\tfrac{u}{2})(2 + \gamma - \log 4\pi) + \sinh(\tfrac{u}{2})(\log \pi - 2u) + A(u) + \sqrt{x} \sum_{n \leq x} \frac{\Lambda(n)}{n} - \frac{\psi(x)}{\sqrt{x}},$$

where Λ is von Mangoldt's function, $\psi(x) = \sum_{n \leq x} \Lambda(n)$, and $A(u) = \frac{1}{2\pi} \int_{\mathbb{R}} H_u(r) \operatorname{Re} \psi_\Gamma(\frac{1}{4} + \frac{ir}{2})\, dr$, with $H_u(t) = (\cos(tu) - \cosh(u/2))/(t^2 + \frac{1}{4})$ and ψ_Γ the digamma function.

Proof. Write $t_\rho = (\rho - \frac{1}{2})/i$, so that $t_\rho^2 + \frac{1}{4} = \rho(1-\rho)$ and $r^{\rho - 1/2} = e^{i t_\rho \log r}$. Since $|r^{\rho-1/2}| \leq \max(r, 1/r)^{1/2}$ and $\sum_\rho 1/|\rho(1-\rho)| < \infty$ [7, Chapter 12], the series converges absolutely, and the symmetry $\rho \mapsto 1 - \rho$ of the zeros gives $K(1/r) = K(r)$. Summed in pairs, $1/(\rho(1-\rho)) = 1/\rho + 1/(1-\rho)$, so $K(1) = 2\sum_\rho 1/\rho = 2 + \gamma - \log 4\pi$ by the Hadamard product [7, Chapter 12].

For the formula let $x \geq 1$ and $u = \log x$. The numerator of $H_u(t) = (\cos(tu) - \cosh(u/2))/(t^2 + \frac{1}{4})$ vanishes at $t = \pm\frac{i}{2}$, so H_u is entire. It is even and real on $\mathbb{R}$, and $|H_u(t)| \leq C/(1 + (\operatorname{Re} t)^2)$ on $|\operatorname{Im} t| \leq 1$. Pairing ρ with $1 - \rho$ gives $\sum_\rho \cos(t_\rho u)/(\rho(1-\rho)) = K(x)$, so $\sum_\rho H_u(t_\rho) = K(x) - \cosh(\frac{u}{2})K(1)$. Weil's explicit formula, Lemma 4.1 below, evaluates the left side, with $g_h(v) = \frac{1}{2\pi} \int_{\mathbb{R}} h(r) \cos(rv)\, dr$:

- $H_u(\pm\frac{i}{2}) = -u \sinh(\frac{u}{2})$, by l'Hôpital's rule;
- from $\frac{1}{2\pi} \int_{\mathbb{R}} \cos(rv)\, dr/(r^2 + \frac{1}{4}) = e^{-|v|/2}$, for $v \geq 0$

$$g_{H_u}(v) = \tfrac{1}{2}\big(e^{-|u-v|/2} + e^{-(u+v)/2}\big) - \cosh(\tfrac{u}{2}) e^{-v/2},$$

which is $-\sinh(\frac{u-v}{2})$ for $v \le u$ and 0 for $v \ge u$. So $g_{H_u}(0) = -\sinh(\frac{u}{2})$, and the prime sum is $2\sum_{n\le x}\Lambda(n)n^{-1/2}\sinh(\frac{1}{2}\log\frac{x}{n}) = \sqrt{x}\sum_{n\le x}\Lambda(n)/n - \psi(x)/\sqrt{x}$;

- the archimedean integral is $A(u)$.

Adding these terms gives the formula. □

The equivalence of RH with $\sum_\rho 1/|\rho|^2 = 2+\gamma-\log 4\pi$ is Burnol's remark [5], also proved by Gun, Murty and Rath [6].

Proposition 2.2. *The Riemann hypothesis holds if and only if $Q_N(a) \ge 0$ for every N and every $a \in \mathbb{Z}^N$, and if and only if $K(r) \ge -K(1)$ for every rational $r > 0$. Under RH there are vectors w_n, $n \ge 1$, in a complex Hilbert space with $\langle w_m, w_n\rangle = G(m,n)$, so that $Q_N(x) = \|\sum_{n\le N} x_n w_n\|^2$.*

Proof. Assume RH. As $|n^\rho| = \sqrt{n}$ and $\sum_\rho 1/|\rho|^2 < \infty$, the vectors $w_n = (n^\rho/|\rho|)_\rho$ lie in ℓ^2 of the zeros, counted with multiplicity. Since $\bar\rho = 1-\rho$ for every zero, $|\rho|^2 = \rho(1-\rho)$ and $\overline{n^\rho} = n^{1-\rho}$, so $\langle w_m, w_n\rangle = \sum_\rho m^\rho n^{1-\rho}/(\rho(1-\rho)) = \sqrt{mn}\,K(m/n) = G(m,n)$, and $Q_N(x) = \|\sum_n x_n w_n\|^2 \ge 0$. Also $|K(r)| \le \sum_\rho |r^{\rho-1/2}|/|\rho|^2 = K(1)$ for every $r > 0$.

Conversely, let $Q_N(a) \ge 0$ for every N and every $a \in \mathbb{Z}^N$. By homogeneity $Q_N \ge 0$ on $\mathbb{Q}^N$, and by continuity on $\mathbb{R}^N$. The kernel is real, as the zeros are symmetric under conjugation, so the 2×2 minor of G at m, n gives $mn\,K(1)^2 \ge mn\,K(m/n)^2$, and in particular $K(r) \ge -K(1)$ for every rational $r > 0$. It remains to derive RH from this lower bound. The series (1) converges locally uniformly, so K is continuous, and $f(u) = K(e^u) + K(1) \ge 0$ for every real u. For $\operatorname{Re} s > \frac{1}{2}$, integrating term by term, as $|e^{u(\rho-1/2)}| \le e^{u/2}$ allows,

$$F(s) = \int_0^\infty f(u)e^{-su}\,du = \frac{K(1)}{s} + \sum_\rho \frac{1}{\rho(1-\rho)(s-\rho+\frac{1}{2})}.$$

The series converges locally uniformly away from the points $\rho - \frac{1}{2}$, so the right side is meromorphic on $\mathbb{C}$, with a pole of residue $m_\rho/(\rho(1-\rho)) \ne 0$ at each point $\rho - \frac{1}{2}$, where m_ρ is the multiplicity of ρ. By Landau's theorem [8, Lemma 15.1], the Laplace transform of a nonnegative function is singular at the real point of its abscissa of convergence, when that is finite. The real singularities of the right side are 0 and the points $\beta - \frac{1}{2}$ for real zeros β, and ζ has no zero in $(0,1)$: there $(1-2^{1-\sigma})\zeta(\sigma) = \sum_{n\ge1}(-1)^{n-1}n^{-\sigma} > 0$ and $1 - 2^{1-\sigma} < 0$. So the abscissa of convergence is at most 0, F is holomorphic on $\operatorname{Re} s > 0$, and $\operatorname{Re}\rho \le \frac{1}{2}$ for every zero. By the symmetry $\rho \mapsto 1-\rho$, every zero has real part $\frac{1}{2}$. □

In Matsumoto and Suzuki's terms [2], K is H_1, and the kernel $K(e^{t-u}) - K(e^t) - K(e^{-u}) + K(1)$ of their screw function is, at the points $\log n$, the Gram matrix of the differences $w_n/\sqrt{n} - w_1$: there their condition is the positivity of Q_N on the hyperplane $\sum_n \sqrt{n}\, x_n = 0$.

Weil's form itself acts on functions. For real $g \in L^2(\mathbb{R})$ vanishing outside a bounded interval, with transform $\hat{g}(z) = \int g(t) e^{izt}\, dt$ and autocorrelation $f(u) = \int g(t) g(t+u)\, dt$, let

$$\mathrm{QW}(g) = 2\hat{g}(\tfrac{i}{2})\hat{g}(-\tfrac{i}{2}) + \big(\operatorname{Re}\psi_\Gamma(\tfrac{1}{4}) - \log\pi\big)\|g\|^2 + \int_0^\infty \big(f(0) - f(u)\big)\frac{e^{u/2}}{\sinh u}\, du - 2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\, f(\log n),$$

Weil's quadratic form in the variable $t = \log u$. No zero of ζ enters it, and for g vanishing outside $[-a, a]$ only the prime powers up to e^{2a} do. Whenever the explicit formula (Lemma 4.1) applies to $h(z) = \hat{g}(z)\hat{g}(-z)$, $\mathrm{QW}(g) = \sum_\rho \hat{g}(t_\rho)\hat{g}(-t_\rho)$, which is $\sum_\rho |\hat{g}(\gamma_\rho)|^2$ under RH. On the hyperplane $\sum_n x_n = 0$ the orbit form is this form on step functions:

Proposition 2.3. *Let $N \ge 2$, let x be real with $\sum_{n\le N} x_n = 0$, and let $g_x(t) = e^{t/2}\sum_{n\le N} x_n 1_{(0,\log n]}(t)$. Then $Q_N(x) = \mathrm{QW}(g_x)$.*

Proof. The transform is $\hat{g}_x(z) = \sum_n x_n \int_0^{\log n} e^{(1/2+iz)u}\, du$, at most a constant over $1 + |\operatorname{Re} z|$ on $|\operatorname{Im} z| \le 1$. So $h(z) = \hat{g}_x(z)\hat{g}_x(-z)$ satisfies the hypotheses of Lemma 4.1, and the right side of the explicit formula is $\mathrm{QW}(g_x)$:

- $h(\pm\frac{i}{2}) = \hat{g}_x(\frac{i}{2})\hat{g}_x(-\frac{i}{2})$;
- $g_h = f$, by Fourier inversion;
- Gauss's integral for ψ_Γ [43, (5.9.12)] and Tonelli turn the archimedean term into $\operatorname{Re}\psi_\Gamma(\frac{1}{4})\|g_x\|^2 + \int_0^\infty (f(0) - f(u))e^{u/2}/\sinh u\, du$.

At t_ρ, $\frac{1}{2} + it_\rho = \rho$, so on the hyperplane $\hat{g}_x(t_\rho) = \sum_n x_n(n^\rho - 1)/\rho = \sum_n x_n n^\rho/\rho$, and likewise $\hat{g}_x(-t_\rho) = \sum_n x_n n^{1-\rho}/(1-\rho)$. Hence $\hat{g}_x(t_\rho)\hat{g}_x(-t_\rho) = \sum_{m,n} x_m x_n \sqrt{mn}\,(m/n)^{\rho-1/2}/(\rho(1-\rho))$, and the sum over the zeros is $\sum_{m,n} x_m x_n \sqrt{mn}\, K(m/n) = Q_N(x)$. □

In $u = e^t$, g_x is $u^{1/2}\sum_n x_n 1_{(1,n]}(u)$, supported in $[1, N]$; on the hyperplane it equals $u^{1/2}\sum_n x_n 1_{(0,n]}(u)$, as the terms on $(0, 1]$ cancel. Weil's form depends on g only through its autocorrelation, so the dilation $u \mapsto u/\sqrt{N}$ changes nothing; it places g_x in the window $[\lambda^{-1}, \lambda]$ with $\lambda = \sqrt{N}$, the windows of Connes and his collaborators [28, 31], and both forms use the prime

powers up to N. So on the hyperplane G_N is Weil's form on the window, compressed to the $N-1$ step functions with jumps at the integers, and the integer vectors of the hyperplane are the integer-valued step functions. Off the hyperplane the same correspondence sends δ_n to $u^{1/2}1_{(0,n]}(u)$, whose transform at t_ρ is n^ρ/ρ, and the first of these, $u^{1/2}1_{(0,1]}$, lies in no window.

Definition 2.4. For $k \geq 1$ let $d_k = \inf_c Q_k(\delta_k - \sum_{n<k} c_n\delta_n)$, the infimum over real $c_1, \ldots, c_{k-1}$; $d_1 = K(1)$. The *pivot form* (A) is the statement $d_k \geq d_1$ for every $k \geq 1$. The *shortest-vector form* (B) is the statement $Q_N(a) \geq d_1$ for every $N \geq 1$ and every $a \in \mathbb{Z}^N \setminus \{0\}$. The *lattice form* (B′) is the statement $Q_N(a) \geq d_1$ for every $N \geq 1$ and every $a \in \mathbb{Z}^N$ with $Q_N(a) \neq 0$.

(B) asks less than (A): under (A), if a_k is the last nonzero coordinate of an integer vector a, then $a_k^2 \geq 1$ and $Q(a) \geq a_k^2 d_k \geq d_k \geq d_1$, by homogeneity. And (B) implies (B′), which in turn makes every Q_N nonnegative at the integer vectors; so by Proposition 2.2 and Corollary 1.2, (B′) too is equivalent to RH. (B) and (B′) differ only at nonzero integer vectors with $Q_N(a) = 0$. Under RH there are none, as G_N is then positive definite by Theorem 1.1; there are some when the orbit vectors are dependent, as for the curves of Section 6. The flat model is the equality case of (A) and (B):

Proposition 2.5. *Let* $K_0(x) = e^{-|\log x|/2} = \frac{1}{2\pi}\int_{\mathbb{R}} \cos(r\log x)/(r^2+\frac{1}{4})\,dr$. *Then* $\sqrt{ab}\,K_0(a/b) = \min(a,b)$ *for* $a, b > 0$, *and vectors with Gram matrix* $c\min(m,n)$ *have every Gram–Schmidt pivot equal to* c.

So replacing the zeros by Lebesgue measure with the weight $1/(\frac{1}{4}+t^2)$ gives the covariance of Brownian motion at integer times, whose increments are orthogonal of equal norm. Burnol reads Weil's positivity in the same light: the covariance of a generalized stochastic process invariant under translation has the form of Weil's functional [5]. In that model (A) and (B) hold with equality, and the increments $\delta_n - \delta_{n-1}$ are all shortest vectors. The orthogonal increments are Lubinsky's orthogonal Dirichlet polynomials for the arctangent density [37], in Dimitrov and Oliveira's form [38, §2.3].

Remark (How the hyperplane sits)**.** For the Gram matrix $\min(m,n)$ the hyperplane $\sum_n x_n = 0$ of Proposition 2.3 is orthogonal to δ_1: it is spanned by the increments $\delta_n - \delta_1$, and $\min(n,1) - \min(1,1) = 0$. In Brownian terms, the increments after time 1 are orthogonal to the value at time 1. With $y_k = \sum_{n\geq k} x_n$, the value of $\sum_n x_n 1_{(0,n]}$ on $(k-1,k]$, this form is $\sum_{k\leq N} y_k^2 = (\sum_n x_n)^2 + \sum_{k\geq 2} y_k^2$, and on the hyperplane it is $\|g_x\|^2$.

For ζ the hyperplane leans towards w_1. Under RH the squared distance from w_1 to it is $1/(\mathbf{1}^{\mathsf{T}} G_N^{-1} \mathbf{1})$, the least value of $Q_N(z)$ over $\sum_n z_n = 1$, that is, of $\sum_\rho |P(\rho)|^2/|\rho|^2$ over the Dirichlet polynomials $P(s) = \sum_{n \le N} z_n n^s$ with $P(0) = 1$. Numerically, in units of d_1, it is

$$0.423, \quad 0.0567, \quad 0.00358, \quad 0.000257 \qquad \text{at } N = 2, 10, 100, 1000,$$

so the angle between w_1 and the hyperplane falls from 40.6° to 0.92°, where the flat model keeps 90°. The pivots of Corollary 1.2 take w_1 first; taken after the hyperplane, w_1 would have only this squared distance as its pivot. Inside the hyperplane, the shortest integer vector for $N \le 30$ is $\delta_2 - \delta_1$, with $Q_N = 4.06\, d_1$, against the bound $c_0 = 1.25\, d_1$ of Corollary 3.2. The unit steps $\delta_k - \delta_{k-1}$, all shortest in the flat model, grow roughly like $\log k$: $Q_N = 22.1\, d_1$ at $k = 10$ and $118.9\, d_1$ at $k = 1000$ (`scripts/window_hyperplane.py`).

3 The main results

We first derive Corollary 1.2 from Theorem 1.1.

Proof of Corollary 1.2. If (A) holds and $x \in \mathbb{R}^N$ is nonzero, let x_k be its last nonzero coordinate. Then $Q_N(x) = Q_k(x) = x_k^2\, Q_k(\delta_k + \sum_{n<k} (x_n/x_k)\delta_n) \ge x_k^2 d_k \ge x_k^2 d_1 > 0$, by homogeneity and the definition of d_k; RH follows by Proposition 2.2. If (B) holds, then $Q_N(a) \ge 0$ at every integer vector, and RH follows by the same proposition. Conversely, assume RH. For $k = 1$, $Q_1(\delta_1) = d_1$. For $k \ge 2$, Theorem 1.1 gives $Q_k(\delta_k - \sum_{n<k} c_n \delta_n) \ge c_0 > d_1$. So (A) holds. If $a \in \mathbb{Z}^N$ is nonzero, either a is supported at 1, and $Q_N(a) = a_1^2 d_1 \ge d_1$, or some a_n, $n \ge 2$, is nonzero, and $Q_N(a) \ge c_0 > d_1$ by Theorem 1.1. So (B) holds. □

Corollary 3.1. *Assume RH. For every $N \ge 1$, in every realisation, the lattice spanned by $w_1, \dots, w_N$ has minimum $\|w_1\|^2 = d_1$, attained at $\pm w_1$ only, and every other nonzero lattice vector has squared length at least $c_0 = 1.2495\ldots d_1$: it is longer than w_1 by a factor at least $\sqrt{1.249} = 1.117\ldots$.*

Proof. The map $a \mapsto \sum_n a_n w_n$ carries $\mathbb{Z}^N$ onto the lattice with $\|\sum_n a_n w_n\|^2 = Q_N(a)$. Let $a \ne 0, \pm\delta_1$. If $a_n \ne 0$ for some $n \ge 2$, then $Q_N(a) \ge c_0$ by Theorem 1.1; otherwise $a = m\delta_1$ with $|m| \ge 2$, and $Q_N(a) = m^2 d_1 \ge 4 d_1 > c_0$. □

Through Proposition 2.3, the lattice gap is a statement about Weil's form on a window:

Corollary 3.2. *Assume RH. Let $N \geq 2$, and let y be an integer-valued function on $(1, N]$, constant on each interval $(k-1, k]$, not identically 0. Then $\mathrm{QW}(u^{1/2}y) \geq c_0 = 0.0577$, the form taken in $t = \log u$.*

Proof. Write $y = \sum_n a_n 1_{(1,n]}$ with $a \in \mathbb{Z}^N$, $a_1 = -\sum_{n\geq 2} a_n$, so that $\sum_n a_n = 0$, $u^{1/2}y = g_a$, and $a_n \neq 0$ for some $n \geq 2$. By Proposition 2.3 and Theorem 1.1, $\mathrm{QW}(g_a) = Q_N(a) \geq c_0$. □

On the whole window $[\lambda^{-1}, \lambda]$, $\lambda = \sqrt{N}$, the least eigenvalue of QW is at most of order $\lambda^9 (\log \lambda)^3 e^{-4\pi\lambda^2}$, by a theorem of Kastoris and of Connes and Kastoris that Andrews, Connes and La Gatta state [31, Theorem 3.1], and numerically of that order [31, Section 4]; a matching lower bound would imply positivity on the window. The integer-valued step functions keep the fixed margin c_0. That bounds the form itself, not its ratio to $\|g\|^2$: large integer vectors near the bottom of the spectrum have Rayleigh quotients near the least one over all step functions, about $2.6/N^2$ at $N = 1000$ (Section 8).

Remark (The pivots are at most logarithmic)**.** Let $N \geq 3$ and $x = N/(N-1)$. Then $d_N \leq U_N = N(d_1^2 - K(x)^2)/d_1$, the least value of $Q_N(\delta_N - c\,\delta_{N-1})$ over real c; under RH, U_N is the squared distance from w_N to the line through w_{N-1}. Moreover

$$U_N = \log \frac{N}{2\pi} + 1 - \gamma + O\Big(\frac{\log^2 N}{N}\Big).$$

For the asymptotic let $u = \log x \leq \log \frac{3}{2}$. No prime power is at most x, so Proposition 2.1 gives $K(x) = \cosh(\frac{u}{2}) d_1 + \sinh(\frac{u}{2})(\log \pi - 2u) + A(u)$. By the asymptotic expansion of ψ_Γ [43, (5.11.2)], $\operatorname{Re} \psi_\Gamma(\frac{1}{4} + \frac{ir}{2}) = \frac{1}{2}\log(r^2 + \frac{1}{4}) - \log 2 + O(1/(1+r^2))$. Split $H_u(r) = (\cos(ru) - 1)/(r^2 + \frac{1}{4}) + (1 - \cosh(\frac{u}{2}))/(r^2 + \frac{1}{4})$ and use $|\cos(ru) - 1| \leq r^2u^2/2$. Every term of $A(u)$ is then $O(u^2)$ except two:

- the constant $-\log 2$, which contributes $\log 2 \sinh(\frac{u}{2})$, since $\int_{\mathbb{R}} H_u = -2\pi \sinh(\frac{u}{2})$;
- $\frac{1}{2\pi} \int_{\mathbb{R}} (\cos(ru) - 1) \log|r|\, dr/r^2 = -\frac{u}{2}(\log \frac{1}{u} + 1 - \gamma)$.

The second uses $\int_0^\infty (1 - \cos v) \log v\, dv/v^2 = \frac{\pi}{2}(1 - \gamma)$, the derivative at $s = 0$ of $\int_0^\infty (1 - \cos v) v^{s-2}\, dv = -\Gamma(s-1)\cos(\frac{\pi}{2}(s-1))$. Hence $d_1 - K(x) = \frac{u}{2}(\log \frac{1}{2\pi u} + 1 - \gamma) + O(u^2)$. With $U_N = 2N(d_1 - K(x)) - N(d_1 - K(x))^2/d_1$ and $u = 1/N + O(1/N^2)$, this gives the asymptotic.

Numerically $U_N = 2.66$, 5.33 and 7.76 at $N = 10^2$, 10^3 and 10^4, and the pivots themselves grow (Section 3.1). So $\log N$ bounds their order, and under RH Theorem 1.1 bounds them below by c_0 at every N.

3.1 Numerical values

Remark. These are computations, not Lean theorems. In ball arithmetic [47] $d_2/d_1 = 1.71841346690985\ldots$ (`scripts/ball_tests/hunt/k2.py`), and $d_k > d_1$ for $2 \le k \le 10^4$, indeed $d_k \ge 2.371\, d_1$ for $3 \le k \le 10^4$, is certified with every rounding bounded, without RH (`scripts/ball_certificate.py`). An independent computation in ball arithmetic up to $k = 500$ puts the minimum of d_k/d_1 at $k = 2$, with $d_{10}/d_1 = 6.35$, $d_{100}/d_1 = 35.67$ and $d_{500}/d_1 = 74.93$ (`scripts/ball_tests/hunt/cert_zeta.py`). So for $2 \le N \le 10^4$ the second minimum of the lattice is $Q_N(\delta_1 + \delta_2)/d_1 = 1.93870544505279\ldots$, certified in ball arithmetic. Indeed an integer vector whose last nonzero coordinate is at $k \ge 3$ has $Q_N(a) \ge d_k \ge 2.371\, d_1$; and on the first two coordinates $Q(a\delta_1 + b\delta_2) = d_1((a+b\omega)^2 + b^2(2-\omega^2))$ with $\omega = G(1,2)/d_1 = -0.5306\ldots$, least at $\pm\delta_1$ and next at $\pm(\delta_1 + \delta_2)$. A Fincke–Pohst enumeration [46] of the integer vectors with $Q_N(a) \le 3d_1$ agrees at $N = 12$ and $N = 30$ (`scripts/ball_tests/shortest/shortest.py`).

4 Proof of the margin

Throughout, $N \ge 2$, x is real, $c_n = x_n\sqrt{n}$, $D(t) = \sum_{n \le N} c_n n^{-it}$, and Λ is von Mangoldt's function. For a test function h let $g_h(u) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\cos(ru)\,dr$.

Lemma 4.1 (Weil's explicit formula)**.** *Let h be even, holomorphic on $|\operatorname{Im} t| \le 1$ with $|h(t)| \le C/(1 + (\operatorname{Re} t)^2)$ there, and real on $\mathbb{R}$. Then, with $t_\rho = (\rho - \frac{1}{2})/i$,*

$$\sum_\rho h(t_\rho) = h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) - g_h(0)\log\pi + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\operatorname{Re}\psi_\Gamma(\tfrac{1}{4} + \tfrac{ir}{2})\,dr - 2\sum_{n \ge 1}\frac{\Lambda(n)}{\sqrt{n}}\, g_h(\log n).$$

This is the explicit formula of Guinand [12] and Weil [1]. Goldston and Gonek state it in this normalisation for h analytic on $|\operatorname{Im} t| \le \frac{1}{2} + \epsilon$ with $|h(t)| \ll (1+|t|)^{-1-\delta}$ [20, Lemma 1], as a specialisation of [21, Theorem 5.12], whose conditions, they note, are that the transform of h is smooth with compact support. None of the test functions here has a smooth transform, so we give the standard proof. The Lean proof, `WeilForm.weilExplicit_zeta`, evaluates the same line integral but takes the sum over the zeros term by

term from the Hadamard product of ξ: each pair of poles of ξ'/ξ, at ρ and $1-\rho$, contributes $2h(t_\rho)$, so it needs no bound on horizontal segments.

Proof. Let $H(s) = h(-i(s-\frac{1}{2}))$, so that $H(\rho) = h(t_\rho)$. It is holomorphic on $-\frac{1}{2} \le \operatorname{Re} s \le \frac{3}{2}$, where $|H(s)| \le C/(1+(\operatorname{Im} s)^2)$, and $H(1-s) = H(s)$ as h is even. The function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ is entire, with zeros exactly at the ρ, and

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi_\Gamma(\tfrac{s}{2}) + \frac{\zeta'}{\zeta}(s).$$

There are heights $T_k \to \infty$ with $\xi'/\xi(\sigma \pm iT_k) \ll \log^2 T_k$ for $-\frac{1}{2} \le \sigma \le \frac{3}{2}$: Davenport proves this bound for ζ'/ζ [7, Chapter 17], the other terms are $\overline{O(\log T_k)}$ by the asymptotic expansion of ψ_Γ [43, (5.11.2)], and $\xi'/\xi(\bar{s}) = \overline{\xi'/\xi(s)}$ gives the heights $-T_k$. Integrate $H\xi'/\xi$ around the rectangle with corners $-\frac{1}{2} \pm iT_k$ and $\frac{3}{2} \pm iT_k$. The horizontal sides contribute $O(T_k^{-2}\log^2 T_k)$, and the substitution $s \mapsto 1-s$, with $\xi(1-s) = \xi(s)$ and $H(1-s) = H(s)$, turns the left side into the right one. As $\sum_\rho |H(\rho)| \le C\sum_\rho 1/(1+\gamma_\rho^2) < \infty$, letting $k \to \infty$ gives

$$\sum_\rho h(t_\rho) = \frac{1}{\pi i}\int_{(3/2)} H(s)\,\frac{\xi'}{\xi}(s)\,ds.$$

Move each term of ξ'/ξ to the line $\operatorname{Re} s = \frac{1}{2}$, where $s = \frac{1}{2} + ir$ and $H(s) = h(r)$; the horizontal segments vanish in the limit each time.

- The term $1/(s-1)$ crosses its pole at $s=1$, with residue $H(1) = h(-\frac{i}{2})$. On the line $\frac{1}{s} + \frac{1}{s-1} = -2ir/(r^2+\frac{1}{4})$ is odd in r, so the first two terms give $2h(-\frac{i}{2}) = h(\frac{i}{2}) + h(-\frac{i}{2})$.
- The constant gives $-\frac{\log\pi}{2\pi}\int_{\mathbb{R}} h = -g_h(0)\log\pi$.
- The digamma term gives $\frac{1}{2\pi}\int_{\mathbb{R}} h(r)\psi_\Gamma(\frac{1}{4}+\frac{ir}{2})\,dr$, whose imaginary part is odd in r.
- On $\operatorname{Re} s = \frac{3}{2}$, $\zeta'/\zeta(s) = -\sum_n \Lambda(n)n^{-s}$ converges absolutely. Moving each term separately, $\frac{1}{\pi i}\int_{(1/2)} H(s)n^{-s}\,ds = \frac{1}{\pi}\int_{\mathbb{R}} h(r)n^{-1/2-ir}\,dr = 2n^{-1/2}g_h(\log n)$. The resulting series converges absolutely, since each term equals its value on $\operatorname{Re} s = \frac{3}{2}$.

The four terms add up to the formula. □

4.1 The test function

Let $\operatorname{sinc} x = (\sin x)/x$, $\mathcal{K}(x) = \operatorname{sinc}^2 x\,(1 + \frac{1}{3}\sin^2 x)$, and

$$\varphi(s) = \frac{999}{1000} \cdot \frac{1 - \mathcal{K}(s/5)}{s^2}, \qquad \hat{\varphi}(v) = \int_{\mathbb{R}} \varphi(s) \cos(sv)\, ds.$$

Lemma 4.2. (i) $\hat{\varphi}(v) = \frac{999}{1000}\pi\big(\frac{25}{9}((\frac{2}{5} - |v|)^+)^3 - \frac{25}{144}((\frac{4}{5} - |v|)^+)^3\big)$*; in particular* $\hat{\varphi}$ *vanishes for* $|v| \geq \frac{4}{5}$ *and* $\hat{\varphi}(0) = \frac{999}{1000} \cdot \frac{4\pi}{45}$. (ii) $0 \leq \varphi(s)$ *and* $\varphi(s)(s^2 + \frac{1}{16}) \leq 1$ *for real* s*, and* $\varphi(s) \leq s^2/3000$ *for* $|s| \leq 5$. (iii) φ *extends to an entire function* $\varphi_{\mathbb{C}}$ *with* $|\varphi_{\mathbb{C}}(z)| \leq C/(1 + (\operatorname{Re} z)^2)$ *on* $|\operatorname{Im} z| \leq 1$*, and* $|\varphi_{\mathbb{C}}(iy)| \leq y^2/3000$ *for* $0 < y \leq \frac{1}{4}$.

Proof. For $\beta > 0$ let $A_\beta(s) = (1 - \operatorname{sinc}^2(\beta s))/s^2$. Since $\operatorname{sinc}^2 2x = \operatorname{sinc}^2 x \cos^2 x = \operatorname{sinc}^2 x\,(1 - \sin^2 x)$, we have $\frac{4}{3}(1 - \operatorname{sinc}^2 x) - \frac{1}{3}(1 - \operatorname{sinc}^2 2x) = 1 - \mathcal{K}(x)$, so $\varphi = \frac{999}{1000}(\frac{4}{3}A_{1/5} - \frac{1}{3}A_{2/5})$.

(i) The triangle $(2 - |w|)^+$ has integral 4 and cosine transform $\int_{-2}^{2}(2 - |w|)\cos(wy)\, dw = 4\sin^2 y/y^2$, so

$$1 - \operatorname{sinc}^2 y = \frac{1}{4}\int_{-2}^{2} (2 - |w|)\big(1 - \cos(wy)\big)\, dw. \tag{3}$$

Fejér's kernel has the transform $\int_{\mathbb{R}}(1 - \cos(as))\cos(sv)\, s^{-2}\, ds = \pi(a - |v|)^+$ for $a \geq 0$. As $0 \leq (1 - \cos(\beta w s))/s^2 \leq \min(2\beta^2, 2/s^2)$ for $|w| \leq 2$, Fubini's theorem applies: $\hat{A}_\beta(v) = \frac{\pi}{4}\int_{-2}^{2}(2 - |w|)(\beta|w| - |v|)^+\, dw$. This vanishes for $|v| \geq 2\beta$, and for $|v| < 2\beta$, with $u = \beta|w|$,

$$\hat{A}_\beta(v) = \frac{\pi}{2\beta^2}\int_{|v|}^{2\beta} (2\beta - u)(u - |v|)\, du = \frac{\pi}{12\beta^2}(2\beta - |v|)^3.$$

At $\beta = \frac{1}{5}$ and $\beta = \frac{2}{5}$ this gives (i), and $\hat{\varphi}(0) = \frac{999}{1000}\pi(\frac{25}{9} \cdot \frac{8}{125} - \frac{25}{144} \cdot \frac{64}{125}) = \frac{999}{1000} \cdot \frac{4\pi}{45}$.

(ii) Clearly $\mathcal{K} \geq 0$. For $\varphi \geq 0$ we show $\mathcal{K} \leq 1$, that is $\sin^2 x\,(1 + \frac{1}{3}\sin^2 x) \leq x^2$. If $x^2 \geq \frac{4}{3}$, the left side is at most $\frac{4}{3}$. If $x^2 < \frac{4}{3}$, let $t = x^2$ and $q = 1 - \frac{t}{6} + \frac{t^2}{120} \in [0, 1]$, so that $|\sin x| \leq |x|q$. As $u \mapsto u(1 + \frac{u}{3})$ increases on $u \geq 0$,

$$\sin^2 x\,(1 + \tfrac{1}{3}\sin^2 x) - x^2 \leq tq^2(1 + \tfrac{1}{3}tq^2) - t = \frac{t^3 P(t)}{622080000},$$

with $P(t) = t^7 - 80t^6 + 2880t^5 - 60800t^4 + 822400t^3 - 7252800t^2 + 39744000t - 110592000$. On $[0, \frac{4}{3}]$, dropping the negative terms other than the constant, $P(t) \leq -110592000 + 39744000 \cdot \frac{4}{3} + 822400(\frac{4}{3})^3 + 2880(\frac{4}{3})^5 + (\frac{4}{3})^7 < 0$.

For $x^2 \le 6$, $|\sin x| \ge |x| - |x|^3/6 \ge 0$, so $\operatorname{sinc}^2 x \ge (1 - \frac{x^2}{6})^2 \ge 1 - \frac{x^2}{3}$; and $1 - \mathcal{K}(x) \le 1 - \operatorname{sinc}^2 x$. Write $\varphi(s)(s^2 + \frac{1}{16}) = \frac{999}{1000}(1 - \mathcal{K}(x))(1 + \frac{1}{16s^2})$ with $x = s/5$. If $s^2 \ge \frac{999}{16}$, then $(1 - \mathcal{K}(x))/(16s^2) \le \frac{1}{999}$, and the product is at most $\frac{999}{1000}(1 + \frac{1}{999}) = 1$. If $s^2 < \frac{999}{16}$, then $x^2 < \frac{5}{2}$, so $\operatorname{sinc}^2 x \ge (\frac{7}{12})^2$ and $1 - \mathcal{K}(x) \le \min(\frac{x^2}{3}, \frac{95}{144})$, and the product is at most $\frac{999}{1000}(\frac{95}{144} + \frac{1}{1200}) < 1$. Finally, for $a = x^2 \le 1$ the bounds $\operatorname{sinc}^2 x \ge (1 - \frac{a}{6})^2$ and $\sin^2 x \ge a(1 - \frac{a}{6})^2$ give

$$1 - \mathcal{K}(x) \le 1 - (1 - \tfrac{a}{6})^2\big(1 + \tfrac{a}{3}(1 - \tfrac{a}{6})^2\big) = \tfrac{7}{36}a^2 - \tfrac{1}{18}a^3 + \tfrac{1}{162}a^4 - \tfrac{1}{3888}a^5 \le \tfrac{7}{36}x^4,$$

so $\varphi(s) \le \frac{999}{1000} \cdot \frac{7}{36} \cdot \frac{s^2}{625} \le \frac{s^2}{3000}$ for $|s| \le 5$.

(iii) Let $H(z) = 1 - \frac{4}{3}\operatorname{sinc}^2(z/5) + \frac{1}{3}\operatorname{sinc}^2(2z/5)$, with sinc extended to an entire function. H is even with $H(0) = 0$, so $\varphi_{\mathbb{C}}(z) = \frac{999}{1000}H(z)/z^2$ is entire, and it extends φ by the identity above. On $|\operatorname{Im} z| \le 1$, $|\sin w| \le e^{|\operatorname{Im} w|}$ gives $|\operatorname{sinc}(z/5)| \le 5e^{1/5}/|z|$ and $|\operatorname{sinc}(2z/5)| \le \frac{5}{2}e^{2/5}/|z|$; so $|\varphi_{\mathbb{C}}(z)| \le C'/|z|^2$ for $|z| \ge 1$, and $\varphi_{\mathbb{C}}$ is bounded on $|z| \le 1$. For the pole, $\operatorname{sinc}(iu) = (\sinh u)/u$, so with $a = y/5$ and $\sigma(a) = (\sinh a)/a$,

$$H(iy) = 1 - \tfrac{4}{3}\sigma(a)^2 + \tfrac{1}{3}\sigma(2a)^2.$$

The exponential series gives $\sigma(a) = p(a) + \epsilon(a)$ with $p(a) = 1 + \frac{a^2}{6} + \frac{a^4}{120}$ and $|\epsilon(a)| \le a^6/4000$ for $|a| \le 1$. The polynomial part is $1 - \frac{4}{3}p(a)^2 + \frac{1}{3}p(2a)^2 = \frac{8}{45}a^4 + \frac{1}{18}a^6 + \frac{7}{1200}a^8$: the terms of order 1 and a^2 cancel. As $|2pe + e^2| \le 5c$ for $0 \le p \le 2$ and $|e| \le c \le 1$, the errors add at most $\frac{4}{3} \cdot \frac{5a^6}{4000} + \frac{1}{3} \cdot \frac{5 \cdot 64a^6}{4000} = \frac{17}{600}a^6$. So for $0 < a \le \frac{1}{20}$, $|H(iy)| \le \frac{8}{45}a^4 + (\frac{1}{18} + \frac{17}{600})a^6 + \frac{7}{1200}a^8 \le \frac{a^4}{5}$, and $|\varphi_{\mathbb{C}}(iy)| = \frac{999}{1000}|H(iy)|/y^2 \le \frac{999}{1000} \cdot \frac{y^2}{3125} \le \frac{y^2}{3000}$ for $0 < y \le \frac{1}{4}$. □

Let $\Phi_N(t) = \varphi(t/N)/N^2$ and $h = |D|^2\Phi_N$. On the real line $h(t) = D(t)D(-t)\Phi_N(t)$, which extends to an even entire function; Lemma 4.1 applies to it, by Lemma 4.2(iii), as $|D(t)| \le N\sum_n |c_n|$ on $|\operatorname{Im} t| \le 1$. For real t and $N \ge 2$, Lemma 4.2(ii) gives $\Phi_N(t) \le 1/(t^2 + N^2/16) \le 1/(\frac{1}{4} + t^2)$, so under RH, where $t_\rho = \gamma_\rho$ is real and $D(\gamma_\rho) = \sum_n x_n n^{1-\rho}$,

$$\sum_\rho h(t_\rho) = \sum_\rho \Big|\sum_{n \le N} x_n n^\rho\Big|^2 \Phi_N(\gamma_\rho) \le Q_N(x). \tag{4}$$

4.2 The prime side

Lemma 4.3. *$2\sum_n \Lambda(n)n^{-1/2}g_h(\log n) = \frac{\hat\varphi(0)}{2\pi N}L(c)$, where $L(c) = 2\sum_{n \le N}\Lambda(n)n^{-1/2}e_n(c)$ is Landau's form and $e_n(c) = \sum_{jn \le N} c_j c_{jn}$.*

Proof. Since c is real, $|D(r)|^2 = \sum_{j,j'\le N} c_j c_{j'} \cos(r\log(j/j'))$, and $\int_{\mathbb{R}} \Phi_N(r)\cos(ru)\,dr = \hat\varphi(Nu)/N$. By $\cos\alpha\cos\beta = \frac12(\cos(\alpha+\beta)+\cos(\alpha-\beta))$,

$$g_h(\log n) = \frac{1}{4\pi N}\sum_{j,j'\le N} c_j c_{j'}\Big(\hat\varphi\big(N\log\tfrac{jn}{j'}\big) + \hat\varphi\big(N\log\tfrac{j}{j'n}\big)\Big).$$

If $jn \ne j'$, the larger of jn and j' exceeds the smaller, which is at most N, by at least 1, so $|\log(jn) - \log j'| \ge \log(1+\frac1N)$; and $\log(1+\frac1N) \ge \frac{2}{2N+1} \ge \frac{4}{5N}$ for $N \ge 2$, by $\log\frac{1+u}{1-u} \ge 2u$ at $u = \frac{1}{2N+1}$. By Lemma 4.2(i) the first term vanishes unless $jn = j'$, and likewise the second unless $j = j'n$. So $g_h(\log n) = \frac{\hat\varphi(0)}{2\pi N} e_n(c)$, which is 0 for $n > N$. □

4.3 The archimedean term

Let $W(t) = \operatorname{Re}\psi_\Gamma(\frac14 + \frac{it}{2}) - \log\pi$ be the archimedean weight. Let $\lambda(s) = \frac{23}{100} + \frac{693}{1000}\sum_{k<30}(1 - \operatorname{sinc}^2(s/(10\cdot 2^k)))$, a staircase whose steps of height $0.693 < \log 2$ at the scales $10\cdot 2^k$ follow $\log s$, and let

$$r(s) = \sum_{j=1}^{9} b_j\Big(\operatorname{sinc}^2\big(\tfrac{s-\tau_j}{2}\big) + \operatorname{sinc}^2\big(\tfrac{s+\tau_j}{2}\big)\Big),$$

with the centres τ_j and weights b_j

τ_j	$\frac32$	2	3	$\frac72$	4	$\frac92$	7	$\frac{15}{2}$	$\frac{17}{2}$
$10^6 b_j$	193	46	1021	593	568	617	1433	78	12

These are nine Fejér bumps, with $\sum_j b_j = 0.004561$, which pay for the heights $s < 12$, where λ lies above $\log(s/2\pi)$. Let $\mu = \varphi\lambda - r$.

Lemma 4.4. (i) $\int \mu(s)\cos(sv)\,ds = 0$ *for* $|v| \ge 1$, *and* $\int \mu \ge \frac78\hat\varphi(0)$. (ii) *For every* $N \ge 2$ *and every real* s, $\mu(s) \le \varphi(s)\big(W(Ns) - \log N\big)$.

Proof. (i) By (3), $\operatorname{sinc}^2(s/A) = \frac14\int_{-2}^{2}(2-|w|)\cos(ws/A)\,dw$, so by Fubini

$$\int_{\mathbb{R}} \varphi(s)\operatorname{sinc}^2(s/A)\cos(sv)\,ds = \frac18\int_{-2}^{2}(2-|w|)\big(\hat\varphi(v+w/A)+\hat\varphi(v-w/A)\big)\,dw.$$

For $A \ge 10$ and $|v| \ge 1$ both arguments have modulus at least $1 - \frac15 = \frac45$, where $\hat\varphi$ vanishes; so the transform of $\varphi\lambda$ vanishes for $|v| \ge 1$. A single kernel has the transform $\int \operatorname{sinc}^2(\frac{s-\tau}{2})\cos(sv)\,ds = 2\pi(1-|v|)^+\cos(\tau v)$, by Fejér's kernel and a translation, and it vanishes for $|v| \ge 1$ too. At $v = 0$

and $A \ge 5$ the arguments w/A lie in $[-\frac{2}{5}, \frac{2}{5}]$, where $\hat\varphi$ is the polynomial of Lemma 4.2(i), and integrating it,

$$\int_{\mathbb{R}} \varphi(s)\big(1 - \operatorname{sinc}^2(s/A)\big)\,ds = \hat\varphi(0)\,\kappa(A), \qquad \kappa(A) = \frac{15}{2A} - \frac{175}{8A^2} + \frac{375}{16A^3}.$$

Each bump, a symmetric pair of kernels, has integral 4π, so

$$\int_{\mathbb{R}} \mu = \hat\varphi(0)\Big(\frac{23}{100} + \frac{693}{1000}\sum_{k<30} \kappa(10\cdot 2^k)\Big) - 4\pi\sum_j b_j = 0.88048\ldots\ \hat\varphi(0),$$

as $\sum_{k<30}\kappa(10\cdot 2^k) = 1.23511\ldots$ and $4\pi\sum_j b_j/\hat\varphi(0) = 0.004561\cdot\frac{45000}{999} = 0.20545\ldots$.

(ii) The functions μ, φ and W are even, and at $s = 0$, where $\varphi(0) = 0$ by Lemma 4.2(ii), the claim is $-r(0) \le 0$. So let $s > 0$.

The weight, uniformly in N. For $\operatorname{Re} w > 0$, $|\psi_\Gamma(w) - \log w + \frac{1}{2w} + \frac{1}{12w^2}| \le \frac{1}{120(\operatorname{Re} w)^4}$. Indeed, by the derivative of Binet's formula [43, (5.9.10)], equivalently [43, (5.9.13)], $\psi_\Gamma(w) = \log w - \frac{1}{2w} - \int_0^\infty \vartheta(v)e^{-wv}\,dv$, where $\vartheta(v) = \frac{1}{e^v-1} - \frac{1}{v} + \frac12 = \sum_{k\ge1}\frac{2v}{v^2+4\pi^2k^2}$ by the partial fractions of coth [43, (4.36.3)], so that

$$0 \le \frac{v}{12} - \vartheta(v) = \sum_{k\ge1}\frac{2v^3}{4\pi^2k^2(v^2+4\pi^2k^2)} \le \frac{v^3}{720};$$

as $\int_0^\infty \frac{v}{12}e^{-wv}\,dv = \frac{1}{12w^2}$, the remainder is at most $\int_0^\infty \frac{v^3}{720}e^{-v\operatorname{Re} w}\,dv = \frac{1}{120(\operatorname{Re} w)^4}$. Apply it to $w = z+2$, $z = \frac14 + ib$, $b > 0$, after two steps of the recurrence $\psi_\Gamma(z) = \psi_\Gamma(z+2) - \frac{1}{z} - \frac{1}{z+1}$. With $R = |w|^2 = \frac{81}{16} + b^2$: $\operatorname{Re}\log w = \frac12\log R \ge \log b$, $\operatorname{Re}\frac{1}{2w} = \frac{9}{8R}$, $\operatorname{Re}\frac{1}{12w^2} = \frac{81/16 - b^2}{12R^2} \le \frac{81}{192R^2}$, $\operatorname{Re}\frac{1}{z} = \frac{1/4}{1/16+b^2}$, $\operatorname{Re}\frac{1}{z+1} = \frac{5/4}{25/16+b^2}$, and $\frac{1}{120(9/4)^4} < \frac{1}{3000}$. So $\operatorname{Re}\psi_\Gamma(\frac14 + \frac{it}{2}) \ge \log\frac{t}{2} - \bar e(t)$ for $t > 0$, where, with $b = t/2$,

$$\bar e(t) = \frac{9}{8R} + \frac{81}{192R^2} + \frac{1/4}{1/16+b^2} + \frac{5/4}{25/16+b^2} + \frac{1}{3000},$$

and $\bar e$ decreases on $[0,\infty)$. At $t = Ns \ge 2s$ this gives, for every $N \ge 2$,

$$W(Ns) - \log N \ge \log\frac{s}{2\pi} - \bar e(Ns) \ge g(s), \qquad g(s) = \log\frac{s}{2\pi} - \bar e(2s).$$

As $\varphi \ge 0$, it suffices that $\varphi(s)(\lambda(s) - g(s)) \le r(s)$, and as $r \ge 0$ this holds wherever $\lambda(s) \le g(s)$.

The bounds.

(a) $1-\operatorname{sinc}^2 x \le u(x) = \min(1, \frac{x^2}{3} - \frac{2x^4}{45} + \frac{x^6}{315})$, from $\operatorname{sinc}^2 x = (1-\cos 2x)/(2x^2)$ and the Taylor bound of cos of degree 8. So $\lambda(s) \le \frac{23}{100} + \frac{693}{1000} S_{30}(s/10)$, with $S_n(x) = \sum_{k<n} u(x/2^k)$. As $u(x) \le x^2/3$, the terms of S_n after the eighth sum to at most $x^2/147456$, so $S_n(x) \le T(x) = S_8(x) + x^2/147456$ for every n; and as $u \le 1$, $S_n(2^j y) \le j - 1 + u(2y) + T(y)$ for $j \ge 1$. Both u and T increase on $[0,\infty)$.

(b) $\operatorname{sinc} x \ge \ell(x) = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \frac{x^6}{5040}$, and ℓ decreases in x^2. On a cell $s \in [s_a, s_b]$ with $\ell_b = \ell(s_b/5) > 0$, $\operatorname{sinc}^2(s/5) \ge \ell_b^2$ and $\sin^2(s/5) \ge (s_a/5)^2 \ell_b^2$, so $\varphi(s) \le \frac{999}{1000}(1 - \ell_b^2(1 + s_a^2 \ell_b^2/75))/s_a^2$. Also $1 - \mathcal{K}(x) \le x^2\theta(x^2)$ for $x^2 \le 6$, where $\theta(a) = \frac{7a}{36} - \frac{a^2}{18} + \frac{a^3}{162} - \frac{a^4}{3888}$ increases on $[0,3]$ and is largest at 3; so $\varphi(s) \le \frac{999}{1000}\theta(\min(s_b^2/25, 3))/25$ if $s_b^2 \le 150$. Let $\varphi_{\mathrm{up}}(s_a, s_b)$ be the smaller bound. Likewise $\operatorname{sinc}^2 z \ge \ell(D/2)^2$ for $|z| \le D/2$ when $\ell(D/2) > 0$; with $|s - \tau_j| \le \max(\tau_j - s_a, s_b - \tau_j)$ and $0 \le s + \tau_j \le s_b + \tau_j$ this bounds r on the cell from below by $r_{\mathrm{lo}}(s_a, s_b)$.

(c) $\log y \ge 2v + \frac{2}{3}v^3$, $v = \frac{y-1}{y+1}$, for $y \ge 1$, applied after removing the powers of 2, with $0.6931471803 < \log 2 < 0.6931471808$ and $\log \pi < 1.1558$. As g increases, this bounds g on $[s_a, \infty)$ from below by $g_{\mathrm{lo}}(s_a)$.

The cells. Each check compares two rational numbers.

- $s \ge 20$. Write $s = 10 \cdot 2^j y$ with $j \ge 1$ and $1 \le y < 2$. By (a), $\lambda(s) \le \frac{23}{100} + \frac{693}{1000}(j + u(2y) + T(y) - 1)$, and $g(s) \ge \log(10y) + j\log 2 - \log 2 - \log \pi - \bar{e}(40y)$. As $\frac{693}{1000} < \log 2$, $\lambda(s) \le g(s)$ follows, for every j at once, from

  $$\tfrac{23}{100} + \tfrac{693}{1000}\big(u(2y) + T(y) - 1\big) \le \log(10y) - \log 2 - \log \pi - \bar{e}(40y),$$

  which is checked on 400 cells of $[1,2)$, with the left side at the right end of a cell and the right side at its left end.

- $12 \le s < 20$. On each of 320 cells, $\frac{23}{100} + \frac{693}{1000} T(s_b/10) \le g_{\mathrm{lo}}(s_a)$, so $\lambda \le g$.

- $\frac{1}{2} \le s < 12$. On each of 460 cells, either $\Delta = \frac{23}{100} + \frac{693}{1000} T(s_b/10) - g_{\mathrm{lo}}(s_a) \le 0$, so that $\lambda \le g$, or $\varphi_{\mathrm{up}}(s_a, s_b)\,\Delta \le r_{\mathrm{lo}}(s_a, s_b)$, with $\ell_b > 0$ and $s_b^2 \le 150$.

- $0 < s \le \frac{1}{2}$. Here $\varphi(s) \le s^2/3000$ by Lemma 4.2(ii), $\lambda(s) \le \frac{23}{100} + \frac{693}{1000} T(\frac{1}{20})$ and $-g(s) \le \log \frac{1}{s} + \log 2 + \log \pi + \bar{e}(0)$. So where $\lambda > g$, $\varphi(\lambda - g) \le \frac{s^2}{3000}(C_0 + \log \frac{1}{s}) \le \frac{1}{3000}(\frac{C_0}{4} + \frac{1}{2})$, with C_0 the sum of the constants, as $s^2 \le \frac{1}{4}$ and $s^2 \log \frac{1}{s} \le s$. The check is $\frac{1}{3000}(\frac{C_0}{4} + \frac{1}{2}) \le r_{\mathrm{lo}}(0, \frac{1}{2})$.

The Lean kernel evaluates the 1181 checks in exact rational arithmetic (`PivotCert`), and `scripts/pivot_constants.py` replays them with Python's rationals. The least margin is 0.0042 in the first two ranges and 2.4% of r_{lo} in the third, near $s = 2.8$, where 455 of the 460 cells need the bumps. □

Lemma 4.5. $h(\frac{i}{2}) + h(-\frac{i}{2}) \geq -\sum_{n\leq N} c_n^2/(6000N)$.

Proof. As h is even, $h(\frac{i}{2}) + h(-\frac{i}{2}) = 2h(\frac{i}{2}) = 2D(\frac{i}{2})D(-\frac{i}{2})\varphi_{\mathbb{C}}(\frac{i}{2N})/N^2$. Since $n^{\pm 1/2} \leq N$ for $n \leq N$, $|D(\pm\frac{i}{2})| \leq N\sum_n |c_n|$, and $(\sum_n |c_n|)^2 \leq N\sum_n c_n^2$. By Lemma 4.2(iii) at $y = \frac{1}{2N} \leq \frac{1}{4}$, $|\varphi_{\mathbb{C}}(\frac{i}{2N})| \leq \frac{1}{12000N^2}$. So $|2h(\frac{i}{2})| \leq 2N^3\sum_n c_n^2/(12000N^4) = \sum_n c_n^2/(6000N)$. □

Proposition 4.6. *Assume RH. For $N \geq 2$ and real x,*

$$Q_N(x) \geq \frac{\hat{\varphi}(0)}{2\pi N}\Big(\big(\log N + \tfrac{7}{8}\big)\sum_{n\leq N} c_n^2 - L(c)\Big) - \frac{1}{6000N}\sum_{n\leq N} c_n^2.$$

Proof. By (4) and Lemma 4.1, as $g_h(0) = \frac{1}{2\pi}\int h$,

$$Q_N(x) \geq h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)W(r)\,dr - 2\sum_{n\geq 1}\frac{\Lambda(n)}{\sqrt{n}}\,g_h(\log n).$$

Substituting $r = Ns$, the integral is $\frac{1}{2\pi N}\int |D(Ns)|^2\varphi(s)(\log N + W(Ns) - \log N)\,ds$, which by Lemma 4.4(ii) is at least $\frac{1}{2\pi N}\int |D(Ns)|^2(\varphi(s)\log N + \mu(s))\,ds$. For $f = \varphi$ and $f = \mu$,

$$\int_{\mathbb{R}} |D(Ns)|^2 f(s)\,ds = \sum_{j,j'\leq N} c_j c_{j'}\hat{f}\big(N\log(j/j')\big) = \hat{f}(0)\sum_{n\leq N} c_n^2,$$

since $\hat{f}$ vanishes for $|v| \geq 1$, by Lemmas 4.2(i) and 4.4(i), and distinct $j, j' \leq N$ have $|\log(j/j')| \geq \log\frac{N}{N-1} \geq \frac{1}{N}$. So the archimedean term is at least $\frac{1}{2\pi N}(\hat{\varphi}(0)\log N + \int\mu)\sum c_n^2 \geq \frac{\hat{\varphi}(0)}{2\pi N}(\log N + \frac{7}{8})\sum c_n^2$, by Lemma 4.4(i). Lemma 4.3 gives the prime side, and Lemma 4.5 the poles. □

4.4 The top vertex

Let $M(y) = \sum_{n\leq y}\Lambda(n)/n$.

Lemma 4.7. *$M(y) \leq \log y + \frac{81}{100}$ for every integer $y \geq 1$. Moreover $M(y) \leq \log y + (\log 3 - 1) + (1 + \log y)/y$ for $1 \leq y \leq 2000$.*

Proof. Let $\psi(y) = \sum_{n\le y}\Lambda(n)$. For an integer $y \ge 1$, $\log y! = \sum_{n\le y}\Lambda(n)\lfloor y/n\rfloor \ge yM(y) - \psi(y)$, and $\log y! \le \int_1^y \log t\,dt + \log y = y\log y - y + 1 + \log y$, so

$$M(y) \le \log y - 1 + \frac{1+\log y}{y} + \frac{\psi(y)}{y}. \tag{5}$$

Now $\psi(y) < 1.03883\,y < y\log 3$ for every $y > 0$ [9, Theorem 12], and with (5) this gives the second inequality, for every $y \ge 1$. It gives the first for $y \ge 3$, as $(1+\log y)/y$ decreases and $\log 3 - 1 + (1+\log 3)/3 < 0.7982$. Finally $M(1) = 0$ and $M(2) = \frac12\log 2$.

The Lean proof does not use [9]. It checks $\operatorname{lcm}(1,\dots,n) \le 3^n$, that is $\psi(n) \le n\log 3$, for $n \le 2000$ by one kernel computation that carries the pair $(\operatorname{lcm}(1,\dots,n), 3^n)$ from n to $n+1$. For $y > 2000$ it uses Chebyshev's bound $\psi(y) \le y\log 4 + 2\sqrt{y}\log y$, which turns (5) into $M(y) \le \log y + \log 4 - 1 + \frac{1+\log y}{y} + \frac{2\log y}{\sqrt{y}}$, below $\log y + 0.767$ for $y \ge 1600$. □

Proposition 4.8. *For every $N \ge 2$, every $d \ge \frac{17}{20}$ and every real c,*

$$\big(\log N + d\big)\sum_{n\le N} c_n^2 - L(c) \ge \tfrac{13}{10}\,c_N^2.$$

Proof. Let c' be c with c_N replaced by 0. A term of $L(c)$ contains c_N only as c_{jn} with $jn = N$, since $j = N$ forces $n = 1$ and $\Lambda(1) = 0$. So, with q running over the divisors $q \ge 2$ of N,

$$L(c) = L(c') + 2c_N \sum_{q\mid N} \frac{\Lambda(q)}{\sqrt{q}}\,c_{N/q}.$$

Schur's test, row by row. For real b, $2\frac{\Lambda(n)}{\sqrt{n}}|b_j b_{jn}| \le b_j^2\frac{\Lambda(n)}{n} + b_{jn}^2\Lambda(n)$. Summing over $jn \le J$, with $\sum_{n\mid m}\Lambda(n) = \log m$, Landau's form L_J on $[1,J]$ satisfies $L_J(b) \le \sum_{j\le J} b_j^2(M(\lfloor J/j\rfloor) + \log j)$. As c' lives on $[1, N-1]$, with $J = N-1$,

$$(\log N + d)\sum_{j<N} c_j^2 - L(c') \ge \sum_{j<N}\sigma_j c_j^2,$$

where $\sigma_j = \log N + d - M(\lfloor (N-1)/j\rfloor) - \log j$; and $\sigma_j \ge d - \frac{81}{100} > 0$, as $M(\lfloor (N-1)/j\rfloor) \le \log\frac{N}{j} + \frac{81}{100}$ by Lemma 4.7.

The parents of the top vertex. At $j = N/q$, $\lfloor (N-1)/j\rfloor = q-1$ and $\log j = \log N - \log q$, so $\sigma_{N/q} = \operatorname{room}_d(q) = d + \log q - M(q-1)$. Completing the square,

$$\operatorname{room}_d(q)\,c_{N/q}^2 - 2c_N\frac{\Lambda(q)}{\sqrt{q}}\,c_{N/q} \ge -c_N^2\frac{\Lambda(q)^2}{q\operatorname{room}_d(q)}.$$

The parents N/q are distinct, and the other rows have $\sigma_j c_j^2 \geq 0$. So, by $\sum_{q|N} \Lambda(q) = \log N$,

$$(\log N + d) \sum_{n \leq N} c_n^2 - L(c) \geq c_N^2 \Big(\log N + d - \sum_{q|N} \frac{\Lambda(q)^2}{q \operatorname{room}_d(q)} \Big)$$
$$= c_N^2 \Big(d + \sum_{q|N} \Lambda(q) \Big(1 - \frac{\Lambda(q)}{q \operatorname{room}_d(q)} \Big) \Big).$$

The numbers. Every term of the last sum is nonnegative, that is $\Lambda(q) \leq q \operatorname{room}_d(q)$: for $q \leq 7$ the rooms are computed from the values of M; for $8 \leq q \leq 2001$ the second bound of Lemma 4.7 gives $\operatorname{room}_d(q) \geq d - (\log 3 - 1) - \frac{1+\log 7}{7} \geq d - 0.52 \geq 0.33$, against $\log q \leq 0.33q$; and for $q > 2001$ the first gives $\operatorname{room}_d(q) \geq d - 0.81 \geq 0.04$, against $\log q \leq 0.04q$. A prime $p \mid N$ contributes at least $\frac{45}{100}$, and the contributions increase with d. At $d = \frac{17}{20}$ the primes $p = 2, 3, 5, 7$ have rooms 1.5431, 1.6020, 1.5734 and 1.5880, and contribute 0.5375, 0.8475, 1.2802 and 1.6053. For $11 \leq p \leq 2001$, $\operatorname{room}_d(p) \geq d - (\log 3 - 1) - \frac{1+\log 10}{10} \geq d - 0.43 \geq 0.42$, and $\log p - (\log p)^2/(0.42p)$ increases in p, from 1.153 at $p = 11$. For $p > 2001$, $\operatorname{room}_d(p) \geq 0.04$, and $\log p - (\log p)^2/(0.04p) \geq 6.88$. Since $N \geq 2$ has a prime factor, the coefficient of c_N^2 is at least $\frac{17}{20} + \frac{45}{100} = \frac{13}{10}$. □

4.5 Proof of Theorem 1.1

Since $\hat{\varphi}(0)/(2\pi) = \frac{999}{1000} \cdot \frac{2}{45}$, the pole term of Proposition 4.6 is exactly $\frac{\hat{\varphi}(0)}{2\pi N} \cdot \frac{5}{1332} \sum c_n^2$. So under RH

$$Q_N(x) \geq \frac{\hat{\varphi}(0)}{2\pi N} \Big(\big(\log N + \tfrac{7}{8} - \tfrac{5}{1332} \big) \sum_{n \leq N} c_n^2 - L(c) \Big),$$

and, as $\frac{7}{8} - \frac{5}{1332} \geq \frac{17}{20}$, Proposition 4.8 gives $Q_N(x) \geq \frac{\hat{\varphi}(0)}{2\pi N} \cdot \frac{13}{10} c_N^2 = \frac{999}{1000} \cdot \frac{2}{45} \cdot \frac{13}{10} x_N^2$, as $c_N^2 = N x_N^2$. If a is an integer vector whose last nonzero coordinate is a_k, $k \geq 2$, then $Q_N(a) = Q_k(a) \geq c_0 a_k^2 \geq c_0$. Finally $c_0/d_1 = 1.2495\ldots$ by (2). □

5 The formalisation

Appendix A names the Lean theorems behind each numbered result, in `lib/` of the repository [50]. The Lean statements of Theorem 1.1 and Corollary 1.2 are

```
theorem pivot_ge_rh (hRH : RiemannHypothesis) {N : ℕ} (hN : 2 ≤ N)
    (x : ℕ → ℝ) :
    999 / 1000 * (2 / 45) * (13 / 10) * x N ^ 2 ≤ BallWeil.form N x

theorem rh_iff_orbitLattice :
    RiemannHypothesis ↔ ∀ (k : ℕ) (c : ℕ → ℝ), 1 ≤ k →
      BallWeil.gram 1 1 ≤
        BallWeil.form k (fun n => if n = k then 1 else -c n)

theorem rh_iff_shortest :
    RiemannHypothesis ↔ ∀ (N : ℕ) (a : ℕ → ℤ),
      (∃ n ∈ Icc 1 N, a n ≠ 0) →
        BallWeil.gram 1 1 ≤ BallWeil.form N (fun n => (a n : ℝ))
```

in the namespace `PivotSharp`, where `RiemannHypothesis` is Mathlib's statement and `BallWeil.gram m n` is $\sqrt{mn}\,K(m/n)$ with K given by the formula of Proposition 2.1: the Gram matrix in the Lean statements is computed from the primes, and does not mention the zeros. `BallWeil.form N x` is $Q_N(x)$. The file `lib/check/Check.lean` prints the axioms of the theorems of Appendix A, and `scripts/check_axioms.py` checks that they are `propext`, `Classical.choice` and `Quot.sound` only. The library's own sources contain no `sorry`, no declared axiom and no `native_decide` (`scripts/check_hygiene.py`). Of the vendored upstream libraries, PrimeNumberTheoremAnd holds two lemmas proved by `sorry`; the axiom check shows that they lie outside the dependency cone of every theorem here. The interval certificate of Lemma 4.4(ii) and the computation $\operatorname{lcm}(1,\dots,n) \le 3^n$ of Lemma 4.7 are checked by kernel evaluation, `decide +kernel`, on exact arithmetic in natural numbers. The toolchain is Lean `v4.35.0-rc3` with Mathlib at commit `0f64d30a`.

The statements that are not Lean theorems are:

- the remarks, which are proved in the text or computed. Among them are the upper bound for the pivots in Section 3, the numerical values of Section 3.1, the whole of Section 6, and the numerics of Section 8;
- the numerical values quoted in the introduction, among them the comparison of densities;
- the equivalence of (B′) with RH after Definition 2.4, which the text derives from Proposition 2.2 and Corollary 1.2;
- the historical and comparative statements of Sections 1 and 7;
- the conjecture and questions of Section 8.

6 The analogues

This section collects tests of (A), (B) and the lattice form (B′) on the natural analogues of the construction. Where the orbit vectors are dependent, (B) would fail at every integral relation among them, so for curves we test (B′). The code for the computations is in `scripts/ball_tests/` of the repository [50], and the limits of the density remark are computed by `scripts/d2_density/`.

6.1 The first step

Remark (The two forms at $k = 2$)**.** Every orbit Gram matrix of this kind has $G(n, n) = nG(1, 1)$, so $G(2, 2) = 2d_1$. Integer vectors see only $\operatorname{Re} G$, since $Q(a) = a^T(\operatorname{Re} G)a$. With $\omega = \operatorname{Re} G(1, 2)/d_1$,

$$Q(a\delta_1 + b\delta_2) = d_1\big((a + b\omega)^2 + b^2(2 - \omega^2)\big),$$

so (B) holds at $N = 2$ exactly when $|\omega| \le 1$; otherwise $\delta_2 - \operatorname{sgn}(\omega)\delta_1$ has $Q = (3 - 2|\omega|)d_1 < d_1$. Since $d_2(\operatorname{Re} G) = (2 - \omega^2)d_1$, (B) at $N = 2$ is (A) at $k = 2$ for $\operatorname{Re} G$, and for a real Gram matrix the two forms coincide on w_1, w_2. For ζ, $\omega = -0.5306\ldots$

6.2 Curves over finite fields

Let $C/\mathbb{F}_q$ have genus g and L-polynomial $\prod_j(1 - \alpha_j T)$, $\alpha_j = \sqrt{q}\, e^{i\theta_j}$. Its zeros are $\frac{1}{2} + i(\theta_j + 2\pi k)/\log q$, $k \in \mathbb{Z}$, and the same formula (1) defines K and G. Poisson summation gives $K(r) = \log q \sum_{k\in\mathbb{Z}} s_k q^{-k/2} e^{-|\log r - k\log q|/2}$, with $s_k = \sum_j \alpha_j^k$ and $s_{-k} = q^{-k}s_k$, so every entry of G is a rational multiple of $\log q$; the term $k = 0$ alone gives $2g \log q \min(m, n)$, the flat model. The computations below use a sample of 148 curves of genus 1 to 3 over $\mathbb{F}_q$, $q \in \{2, 3, 4, 5, 7, 9, 11, 13\}$: smooth hyperelliptic models, and the curves with extremal L-polynomials, namely the maximal and minimal elliptic curves over $\mathbb{F}_4$ and $\mathbb{F}_9$ and the Hermitian quartic over $\mathbb{F}_9$.

Remark (The null vector is integral)**.** Let $m(z) = \sum_{k\le d} m_k z^{d-k}$ be the minimal polynomial of Frobenius, the squarefree part of $L^*(z) = z^{2g}L(1/z) = \prod_j(z - \alpha_j)$: it is monic with integer coefficients, of degree $d \le 2g$. Then $\sum_k m_k w_{q^{d-k}} = 0$. Indeed $q^{i\gamma_\rho} = e^{i\theta_j}$ depends only on the class j of the zero, so $w_{q^a} = \sum_j \alpha_j^a P_j w_1$, where P_j projects onto the zeros of class j, and $\sum_k m_k w_{q^{d-k}} = \sum_j m(\alpha_j)P_j w_1 = 0$. So the pivot at index q^d vanishes, and (A) fails for every curve. So does (B), at the integer vector with entry m_k

at $n = q^{d-k}$, which has $Q = 0$; but that vector is the relation itself, the zero vector of the lattice, which (B′) does not see. For the 148 curves of the sample this was checked in exact rational arithmetic; q^d runs from 4 to 13^6.

Remark (Below the null vector)**.** On $n \le \min(20, q^d - 1)$, (A) fails for 59 of the 148 curves, 43 of them at $k = 2$, and (B′) fails for exactly those 43, all at $N = 2$; on the first 20 indices not divisible by q, where no null vector is forced, (A) fails for 53 and (B′) for 39, again only at the first two indices. Failures were certified in exact rational arithmetic, passes by a Fincke–Pohst search.

6.3 The Frobenius orbit

The geometric analogue is the orbit of the graphs Γ^k of the powers of Frobenius on $C \times C$. Hallouin and Perret compute its Gram matrix, projected to the part orthogonal to the fibres: $2gq^k$ on the diagonal and $q^{\min(i,k)} s_{|i-k|}$ off it [39, Lemma 3], and read Weil's bound as a 2×2 Gram determinant and Ihara's as a 3×3 one [39]. In the variable z, the orbit is $1, z, z^2, \dots$ with the form $T(P) = \sum_j |P(\alpha_j)|^2$.

Remark (The forms part)**.** (A) fails for every curve: the orbit vectors $z^0, \dots, z^d$ are dependent, so the pivot at z^d is 0; already $d_2 \ge d_1$ means $|q+1-\#C(\mathbb{F}_q)| \le 2g\sqrt{q-1}$, which maximal curves violate. But the integer span of the orbit is the lattice $\mathbb{Z}[z]/(m)$, since the dependencies are integral, and T is an integral form with $T(1) = 2g$. If m is irreducible, then $L^* = m^e$, and $\prod_j P(\alpha_j) = \mathrm{Res}(m, P)^e$ is a nonzero integer for $P \notin (m)$, so by the arithmetic–geometric mean inequality $T(P) \ge 2g$: (B′) holds. Equality at some $P \ne \pm 1$ forces $P(\alpha)$ to be a root of unity other than ± 1, as at $\alpha = 1+i$ over $\mathbb{F}_2$, where $P = z - 1$. In the sample, m is irreducible for 132 curves, and an exact search finds no failure of (B′) for the other 16.

6.4 Dirichlet characters

For a primitive character χ mod q of parity a, the same formula over the zeros of $L(s, \chi)$ defines K_χ and G, and the explicit formula gives a closed form [7, Chapter 19]. With $K_1 = K_\chi(1)$ and $\ell = L'/L(1, \chi)$, $G(1, 2) = K_1 + 2\tau_a + \ell$, where $\tau_0 = -\frac{1}{4}\log\frac{27}{4}$ and $\tau_1 = \frac{3}{4}\log\frac{3}{4}$.

Remark (The criterion at $k = 2$)**.** Since $K_1 = \log\frac{q}{\pi} + \psi_\Gamma(\frac{1+a}{2}) + 2\,\mathrm{Re}\,\ell$, (B) holds at $N = 2$ if and only if $\frac{1}{5}L_a \le K_1 \le L_a$, with $L_a = \log\frac{q}{\pi} + \psi_\Gamma(\frac{1+a}{2}) - 4\tau_a$, for every χ; for real χ this is also (A) at $k = 2$. The upper end says that the low zeros carry more weight than their density accounts for.

Remark (Density)**.** Let $X = -\sum_p (\log p)\,\epsilon_p/(p-\epsilon_p)$ be Lamzouri's random model for $L'/L(1,\chi_D)$ [42], with independent ϵ_p equal to 1 and to -1 with probability $p/(2(p+1))$ each and to 0 with probability $1/(p+1)$. For real χ_D, ℓ is real. Put $\nu_a = \log\frac{q}{\pi} + \psi_\Gamma(\frac{1+a}{2})$, so that $K_1 = \nu_a + 2\ell$ and $\nu_a = \log|D| + O(1)$. By the criterion at $k=2$, $d_2 < d_1$ exactly when $\ell > -2\tau_a$ or $\ell < -\frac{2}{5}(\nu_a + \tau_a)$, where $-2\tau_0 = \frac{1}{2}\log\frac{27}{4} = 0.954\ldots$ and $-2\tau_1 = \frac{3}{2}\log\frac{4}{3} = 0.431\ldots$. So $d_2 < d_1$ whenever $\ell > \frac{1}{2}\log\frac{27}{4}$, for either parity, and then χ_D fails both forms.

Lamzouri shows that the moments of ℓ over the fundamental discriminants $|D| \le x$, leaving out the $O(\log x)$ whose L-function has a Siegel zero, tend to those of X [42, Theorem 1.6], and that so do their Laplace transforms, outside $O(x^\varepsilon)$ discriminants [42, Theorem 1.5]; so the distribution of ℓ tends to that of X. His proof approximates the powers of ℓ by Dirichlet polynomials, for all but $O(x^{3/4})$ discriminants, by Heath-Brown's zero-density estimate for real characters, and then averages $\chi_D(n)$: for $n = m^2$ by counting the D prime to m, and for the other n by the Pólya–Vinogradov inequality [42, Section 2]. The distribution of X has no atoms: by Lévy's theorem [44], a convergent series of independent discrete random variables has an atom only if the product over its terms of their largest atoms is positive, and the term at p has largest atom $p/(2(p+1)) < \frac{1}{2}$. So the proportion of fundamental discriminants $|D| \le x$ with $d_2 < d_1$ is at least $P(X > \frac{1}{2}\log\frac{27}{4}) + o(1) = 0.0511\ldots + o(1)$.

Within one sign of D the same argument should give the exact limits; we only sketch it. The approximation is made D by D, and both averages should hold over the D of one sign, with the same proportions. If so, the proportion of $D > 1$ with $d_2 < d_1$ tends to $P(X > \frac{1}{2}\log\frac{27}{4}) = 0.0511\ldots$, and that of $D < 0$ to $P(X > \frac{3}{2}\log\frac{4}{3}) = 0.1609\ldots$, as the proportion with $\ell < -\frac{2}{5}\log|D| + O(1)$ tends to 0.

Remark (The scans)**.** Of the 29468 primitive characters with $q \le 400$, 5983 fail (A) at $k=2$ and 5443 fail (B), all at $N=2$, on $n \le 12$; the other 540 fail (A) only through $\operatorname{Im}\ell$. The four that first fail (A) at $k=3$, two mod 13 and two mod 61, pass (B). Of the 6086 real primitive characters with $q \le 10^4$, (B) fails for exactly the 637 with $d_2 < d_1$. Failures were certified in ball arithmetic.

6.5 Li's basis

Remark. Lagarias's Theorem 3.1 [10] gives the Gram matrix of Weil's scalar product on Li's functions $\mathcal{G}_n(s) = 1 - (1-1/s)^n$: $\langle \mathcal{G}_n, \mathcal{G}_m\rangle_W = \lambda_n + \lambda_{-m} -$

λ_{n-m}, with Li's coefficients λ_n [11]. With $F_n(s) = n^s/s$, the same pairing gives $G(m,n)$, and $\mathcal{G}_1 = F_1 = 1/s$: the two are Gram matrices of one scalar product on two orbits of one vector. Under RH, $z_\rho = 1 - 1/\rho$ lies on the unit circle and the pivots of Lagarias's matrix are the squared norms of the monic orthogonal polynomials for $\sum_\rho \delta_{z_\rho}/|\rho|^2$. They cannot increase, and they tend to 0 by Szegő's theorem [45, Chapter 2]: numerically 0.0462, $7.42 \cdot 10^{-5}$, ..., $5.89 \cdot 10^{-23}$ at $k = 8$ (`scripts/ball_tests/prior_art/li_orbit.py`). In the flat model both orbits have constant pivots; with the zeros of ζ the dilation orbit's pivots rise and the shift orbit's fall.

7 Related work

The positivity layer. Weil's criterion [1] in Bombieri's form [25] is the starting point. The kernel K is Matsumoto and Suzuki's H_1 [2], and their Theorem 1.3 is a continuous analogue of Proposition 2.2; see also Suzuki's study of the screw function [27]. Nicolas notes the 2×2 layer under RH [4]. The value of d_1 is classical [7, Chapter 12], and the equivalence of RH with $\sum_\rho 1/|\rho|^2 = d_1$ is Burnol's remark (Section 2).

The method. Weil's explicit formula with band-limited test functions has been a standard tool since Montgomery's work on pair correlation [16]. Goldston and Gonek [20] use it with the band-limited majorants and minorants of Beurling and Selberg [22], and Carneiro, Milinovich and Soundararajan [23] choose its test functions by Fourier optimisation. Mean values of $|A(\gamma_\rho)|^2$ over the zeros, for Dirichlet polynomials A, bound the gaps between zeros in the work of Montgomery and Odlyzko [17] and of Conrey, Ghosh and Gonek [18]. Their prime terms are Landau's form weighted by the transform of a short window, in the limit of large height. The coefficients are chosen to optimise it, and Conrey, Ghosh and Gonek also bound it for all coefficients at once: so the method cannot exhibit gaps shorter than half the mean spacing, or longer than 6.2 times it, as Bui, Milinovich and Ng report [19, Section 5]. In Theorem 1.1 the transform of the test function is supported in $[-\frac{4}{5}, \frac{4}{5}]$, so the weight is the constant $\hat{\varphi}(0)$ and the identity is exact at every N, with no limit in the height; and Schur's test at the top vertex of the divisibility graph bounds Landau's form by the diagonal with $\frac{13}{10} c_N^2$ to spare, which is what makes the margin.

Weil's form on finite-dimensional spaces. Yoshida [24] localises Weil's criterion to windows $[-a, a]$ of the variable $\log u$: RH holds if and only if Weil's form is non-degenerate on a completed space $\hat{K}(a)$ of window functions for every $a > 0$ [24, Theorem 2]. He proves the form non-degenerate

on the smooth functions supported in each window, and, for $\mathbb{Q}$, on the space $K(a)$ of restrictions of smooth periodic functions [24, Propositions 2 and 7]; positive on small windows [24, Lemma 2]; and, on the subspaces of finite codimension where the low Fourier modes vanish, at least any prescribed multiple of the squared L^2 norm [24, Lemma 3]. He also orthogonalises the high Fourier modes by Gram–Schmidt. In his setting too, only the hyperplane $\sum_n x_n = 0$ of the orbit's span fits a window. Bombieri [25, 26] studies the form's minimisation over test functions on windows. Connes and Consani explain, through a semi-local trace formula, the positivity of its archimedean part for test functions supported in $(\frac{1}{2}, 2)$, which Yoshida had proved by an explicit computation [28], and find its small eigenvalues on $[\lambda^{-1}, \lambda]$ extremely small [29]; Suzuki [30] places these results in the frame of the screw function. Groskin [32] shows that every value of a finite Galerkin truncation of Weil's form is an exact sum over the zeros, for a band-limited test function, and certifies the positivity of such truncations by an interval LDL^{T} factorisation, that is, by the signs of their pivots. Closest in aim, Andrews, Connes and La Gatta [31] find numerically that conditioning Weil's form on $[\lambda^{-1}, \lambda]$ by prolate vectors and the pole functional removes its minuscule spectrum. Under a hypothesis they prove that the form is bounded below by an explicit positive constant on the orthogonal complement of an explicit finite-dimensional space, a margin of the shape of Yoshida's Lemma 3; under further estimates, one of which they leave open, this reduces its positivity to a two-sided bound on a Schur complement on a finite block [31, §8]. That margin holds away from a finite-dimensional space of test functions on a window, and it serves a proof of positivity. The margin here bounds every Schur complement along one basis, the orbit, and it is equivalent to RH. The two settings meet exactly: on the hyperplane $\sum_n x_n = 0$, the orbit form is their form on the window with $\lambda = \sqrt{N}$, compressed to step functions (Proposition 2.3), where the margin becomes a lower bound c_0 at the integer-valued step functions (Corollary 3.2). None of these works bounds pivots below by a first pivot or considers integer lattices.

Distances and extremal problems. In the Nyman–Beurling criterion, in Báez-Duarte's form [34], the squared distance in $L^2(0, \infty)$ from $1_{(0,1]}$ to the span of the dilates $x \mapsto \{1/(nx)\}$, $n \le N$, of the fractional part of $1/x$ tends to 0 if and only if RH holds. Báez-Duarte, Balazard, Landreau and Saias [33] bound it below by about $S/\log N$, where S is the sum of $1/|\rho|^2$ over the distinct zeros, which is d_1 when RH holds and the zeros are simple; Burnol [35] counts multiple zeros with their multiplicities squared, and Bettin, Conrey and Farmer [36] show, conditionally, that this is its size. Here the squared distances of the w_k from the span of the earlier ones are bounded below by

d_1 instead. Dimitrov and Oliveira solve a dual extremal problem [38], whose constants, under RH, are partial sums of the series $\sum_\rho 1/|\rho|^2 = d_1$.

Analogues. Hallouin and Perret's Gram matrix of the Frobenius orbit [39] is the function-field analogue of G_N (Section 6). Conrey and Li [40] show that de Branges's positivity conditions, stronger than RH, fail for ζ. The margin here behaves the other way round: for ζ it holds exactly when RH does, while its pivot form fails for every curve over a finite field, where RH is a theorem, and both forms fail for a positive proportion of real characters.

Li's criterion. Li's coefficients [11] are values of Weil's form in another basis (Section 6.5). According to its review in zbMATH (Zbl 1326.11047), which is all we have read of it, Mazhouda [41] proves that GRH for $L(s,\chi)$, χ primitive of conductor $q > 1$, holds if and only if Li's coefficients $\lambda_\chi(n)$ increase with n. That would give, under GRH, the margin $\lambda_\chi(n) \geq \lambda_\chi(1) > 0$ in Li's criterion: a margin of another kind, on the diagonal of Li's basis, where the pivots collapse, and one that excludes ζ.

8 The smallest eigenvalue, and questions

Under RH the pivots of G_N stay above c_0, but its smallest eigenvalue $\lambda_N = \min_{\|x\|=1} Q_N(x)$ appears to tend to 0: the lattice keeps a uniform gap at the integer points while the real quadratic form degenerates.

Remark. These are computations, not Lean theorems. They use G_N computed from the primes by Proposition 2.1, in double precision.

- Cholesky factorisation and block inverse iteration give $\lambda_N N$ at $N = 10^3,\ 2\cdot10^3,\ 4\cdot10^3,\ 8\cdot10^3,\ 1.2\cdot10^4,\ 1.6\cdot10^4,\ 2\cdot10^4,\ 2.5\cdot10^4$ and $3\cdot10^4$ (`scripts/lambda_large.py`). The least-squares line through these nine values, $\lambda_N N = 0.04335\log N + 0.0322$, fits them to within $2\cdot10^{-4}$. Over the same range $\lambda_N N/\log N$ falls from 0.0480 to 0.0465, so a limit near 0.043 is an extrapolation, not a computed value; and it depends on the law fitted, as a power $0.056(\log N)^{0.92}$ fits the same values to within $6\cdot10^{-4}$.

- The alternating vector $x_n = (-1)^{n+1}/n$ shows the mechanism. Its polynomial $\sum_{n\leq N} x_n n^\rho$ is a partial sum of the alternating series for $\eta(1-\rho) = (1-2^\rho)\zeta(1-\rho)$, which vanishes at every zero. So the polynomial is minus a tail of that series, small at the zeros of moderate height. Its Rayleigh quotient is $0.152\log N/N$ for $10^2 \leq N \leq 3\cdot10^4$ (`scripts/alt_vector.py`).

- On the hyperplane $\sum_n x_n = 0$, the least value of $\mathrm{QW}(g_x)/\|g_x\|^2 = Q_N(x)/\sum_{k\geq 2} y_k^2$, in the notation of the remark at the end of Section 2, times N^2 is 1.39, 1.67, 1.98, 2.26 and 2.56 at $N = 10$, 30, 100, 300 and 1000 (`scripts/window_hyperplane.py`): of order $1/N^2$, up to a factor that grows slowly, roughly like $\sqrt{\log N}$ over this range. On the whole window the least eigenvalue is far smaller (Section 3). So the step functions of the orbit stay far from the nearly null, prolate directions of the window.

Conjecture 1. $\lambda_N \sim c \log N/N$ for a constant $c > 0$; the fit above suggests $c \approx 0.043$.

Remark (Questions)**.** (1) Is the second minimum of the lattice attained at $\pm(\delta_1 + \delta_2)$ for every N, beyond the $N \leq 10^4$ of Section 3.1? It would follow from $d_k \geq 1.939\, d_1$ for every $k \geq 3$. (2) Is there an integral structure for ζ that explains (B), as integrality explains (B′) for the Frobenius orbit? (3) For Dirichlet L-functions, does a bounded number of pivots decide (A)?

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A The claims table

The table names the Lean theorem behind each numbered result. All are in `lib/core`. A name's first component is its namespace, which is also the name of its file, except that `WeilForm.weilExplicit_zeta` is in `WeilZeta.lean`.

Claim	Lean
Theorem 1.1: $Q_N(x) \geq c_0 x_N^2$	`PivotSharp.pivot_ge_rh`
the integer vectors	`PivotSharp.lattice_gap`
Corollary 1.2: RH $\iff$ (A)	`PivotSharp.rh_iff_orbitLattice`
(A) $\Rightarrow$ RH	`BallWeil.rh_of_pivots`
RH $\Rightarrow$ (A)	`PivotSharp.orbitLattice_of_rh`
Corollary 1.2: RH $\iff$ (B)	`PivotSharp.rh_iff_shortest`
RH $\Rightarrow$ (B)	`PivotSharp.shortest_of_rh`

Claim	Lean
Corollary 3.1	`PivotSharp.shortest_unique`, `PivotSharp.lattice_second`
Corollary 3.2	`OrbitWindow.weilQg_step_int_ge`
Proposition 2.1	`BallHadamard.summable_hw`, `BallHadamard.kerH_one`, `BallHadamard.kerH_inv`, `BallPsi.kerH_explicit`
Proposition 2.2	`BallWeil.rh_iff_form`, `BallWeil.rh_iff_weil_bound` (the lower bound, stated for $m+n-G(m,n)$), `BallHadamard.rh_iff_realizationH`, `BallWeil.form_eq_norm_sq`
Proposition 2.3	`OrbitWindow.form_eq_weilQg`, `OrbitWindow.weilQg_hasSum`
Proposition 2.5	`BallFlat.kerFlat_eq_integral`, `BallFlat.gram_flat`, `BallFlat.flat_pivot`, `BallFlat.inner_increments`
Lemma 4.1	`WeilForm.weilExplicit_zeta`
Lemma 4.2	`PivotPhi.integral_phi_cos`, `PivotPhi.phi_nonneg`, `PivotPhi.phi_mul_le`, `PivotPhi.phi_le_sq`, `PivotPhi.phiE_strip`, `PivotPhi.norm_phiE_I_le`
Lemma 4.3	`PivotSharp.prime_eq`
Lemma 4.4(i)	`PivotMinorFT.integral_mu_cos_eq_zero`, `PivotMinorFT.integral_mu_ge`
Lemma 4.4(ii)	`PivotCert.mu_le`
Lemma 4.5	`PivotSharp.pole_ge`
Proposition 4.6	`PivotSharp.form_ge`, whose hypothesis `PivotSharp.MuLe` is `PivotSharp.muLe`
Lemma 4.7	`PivotTop.mertens_le_uniform`, `PivotTop.mertens_le_small`
Proposition 4.8	`PivotTop.top_bound`

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